



**SCHOOL OF COMPUTER SCIENCES
UNIVERSITI SAINS MALAYSIA**

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Assignment 2 Report

Toxland VR: Hop & Help

Meta Quest to PICO 4 Conversion Project

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1 Abstract

Toxland VR: Hop & Help is an educational Virtual Reality application that teaches young children to recognise and safely handle toxic or hazardous household materials. Inspired by the classic Snakes and Ladders board game, players take on the role of Junior Safety Rangers and progress across a giant interactive board representing the rooms of a home, making safety decisions about items such as cleaning chemicals, medicines, batteries, detergents, pesticides, and aerosol sprays. Correct decisions advance players via Good Practice ladders, while unsafe choices trigger Toxic Trap penalties accompanied by educational explanations. This project converts and adapts an existing Unity3D VR build developed with the Meta SDK for the Meta Quest headset into an XR-compatible build that runs on the PICO 4 headset using the PICO XR SDK. The conversion covers Unity XR settings, controller input mapping, camera rig and locomotion, UI interaction, and Android build configuration. This report documents the VR concept, background study, conversion methodology, system development, and testing evidence confirming that the application runs successfully on the target platform.

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2 VR Concept

This section presents the conceptual design of Toxland VR: Hop & Help, including its theme, intended audience, world design, storyline, storyboard, and team structure.

2.1 Concept Theme

The concept theme of Toxland VR is household hazard safety education, delivered through a board-game-inspired progression system. The game is a hybrid of an educational and entertainment-focused experience: it uses the familiar mechanics of Snakes and Ladders — “Good Practice” ladders that reward safe behaviour and “Toxic Trap” penalties that punish unsafe behaviour — to reinforce real-world household safety habits around cleaning chemicals, medicines, batteries, detergents, pesticides, and aerosol sprays.

2.2 Target Audience

The primary target audience is young children **[TODO: confirm exact target age range, e.g. 8–12 years, and justify the choice based on cognitive development and reading-level suitability of the safety content]**. The VR medium is chosen because immersive, first-person decision-making is expected to reinforce safety habits more effectively than passive instruction such as posters or textbook lessons, while the game-like reward/penalty structure keeps engagement age-appropriate.

2.3 World Design

Players enter the colourful world of Toxland, where everyday household items come to life. The board is divided into rooms representing different parts of a home:

- Kitchen
- Bathroom
- Laundry area
- Garden
- Storage room

Each room hosts Good Practice Spaces, Bad Practice Spaces, and Learning Card spaces relevant to the hazards typically found in that part of the house **[TODO: insert environment concept art / world map screenshot once the 3D environment is built]**.

2.4 Storyline

Players take on the role of Junior Safety Rangers tasked with travelling across the Toxland board to reach the Safety Champion Zone. Along the way they:

1. Roll a virtual dice or teleport to the next board space.
2. Encounter a household safety scenario at that space.
3. Choose a safe or unsafe action via VR controller selection.
4. Advance via a Good Practice ladder (safe choice) or fall back via a Toxic Trap (unsafe choice), with an educational explanation shown either way.
5. Occasionally land on a Learning Card space, triggering a safety tip, fun fact, mini quiz, or hazard identification challenge.

The game ends when a player reaches the Safety Champion Zone, at which point the final score is tallied and displayed on the end screen.

2.5 Storyboard

[TODO: insert a 4–6 frame storyboard illustrating: (1) the welcome screen, (2) entering the Toxland board, (3) a Good Practice / Bad Practice decision moment, (4) a Learning Card quiz, and (5) reaching the Safety Champion Zone / end screen]

2.6 Team Structure

Table 1: Project team structure and responsibilities

Name	Role	Responsibility
Lai Yicheng	[TODO: role]	Project planning and management
Koay Chun Keat	[TODO: role]	Unity scripting, game logic, scoring system
Yong Pei Yi, Angeline	[TODO: role]	Board design, household objects, environment
Tan Ke Jun	[TODO: role]	Character and object animations
Tan Yi Pin	[TODO: role]	Menus, educational interfaces, feedback systems

[TODO: confirm and assign each member to one of: Project Director, Programmer, 3D Artist, Animator, UI/UX Designer]

3 Background Study

3.1 Existing Applications

Table 2 compares Toxland VR against existing child-safety and hazard-awareness educational VR/serious-game applications.

Table 2: Comparison with existing child safety / hazard awareness applications

Application	Purpose	Platform	Interaction Model
[TODO: Application 1 name]	[TODO: educational objective]	[TODO: platform]	[TODO: interaction method]
[TODO: Application 2 name]	[TODO: educational objective]	[TODO: platform]	[TODO: interaction method]
[TODO: Application 3 name]	[TODO: educational objective]	[TODO: platform]	[TODO: interaction method]

[TODO: Research and cite 2–3 verifiable existing educational games or VR applications related to child safety education, hazard awareness, interactive learning, or serious games. Compare their educational objectives, interaction methods, user engagement, and learning effectiveness against Toxland VR. Add corresponding entries to references.bib and cite them here.]

3.2 Key Innovation

Toxland VR builds on the reviewed applications by combining a familiar board-game progression mechanic (Snakes and Ladders) with immersive VR decision-making, rather than relying on passive narration or 2D interfaces. The project’s specific technical innovation is its cross-headset portability: the same gameplay and educational content, originally built against the Meta SDK for Meta Quest, is re-engineered to run on the PICO XR SDK for the PICO 4 headset, demonstrating that the educational design is not locked to a single hardware ecosystem [TODO: expand with any additional design innovations introduced during development, e.g. adaptive difficulty, multiplayer, localisation].

4 Methodology

4.1 Platforms

Toxland VR is developed in Unity3D. The original source code targets the Meta Quest headset using the Meta SDK (Meta XR All-in-One SDK / OVR components). This project replaces the Meta Quest-specific dependencies with the PICO XR SDK, while retaining Unity's XR Interaction Toolkit and OpenXR backend where possible to maximise cross-platform compatibility [**TODO: confirm exact Unity version, XR Interaction Toolkit version, and OpenXR vs. proprietary PICO SDK usage**].

4.2 Minimum Specification Requirements

Table 3: Minimum system requirements

Requirement	Specification
Target headset	PICO 4
Operating system	Android 10+
Storage	1 GB
Input	Joypad / crosshair (PICO controller or gaze-based selection)

4.3 System Architecture

[**TODO: insert a flowchart or block diagram of the overall game flow (welcome screen → board loop → end screen) and the high-level Unity scene/manager architecture (e.g. GameManager, BoardManager, ScoreManager, UIManager)**]

4.4 Game Flow

1. **Welcome Screen** — player is introduced to Toxland and the Junior Safety Ranger role.

2. **Board Loop** — player rolls the dice or teleports to the next space, the space's event (Good Practice, Bad Practice, or Learning Card) is resolved, and the score is updated.
3. **Safety Champion Zone** — reached when the player completes the board, ending the loop.
4. **End Screen** — final score and a summary of safety lessons learnt are displayed.

4.5 Interactive Zone & Triggers

- **Dice/teleport trigger** — initiates player movement to the next board space.
- **Good Practice Space trigger** — detects landing on a ladder space and grants bonus movement.
- **Bad Practice Space trigger** — detects landing on a toxic trap space, applies a movement penalty, and displays an educational warning.
- **Learning Card trigger** — spawns a safety tip, fun fact, mini quiz, or hazard identification challenge UI panel.
- **Score trigger** — updates the score system on correct safety decisions, quiz answers, hazard identification, and game completion.

5 System Development

5.1 Navigation Control

Players move along the virtual board using joystick-based locomotion, teleportation to board spaces, and dice-roll interaction. The original Meta Quest input mapping (Meta XR controller bindings) is reconfigured for PICO 4 controllers via the PICO XR SDK's input action mappings **[TODO: document the specific Meta Quest input components that were replaced and the equivalent PICO XR input actions used]**.

5.2 Interaction System

VR controller selection (grab, point-and-select, crosshair) is used for choosing safety actions and interacting with household objects. Interaction components originally bound to the Meta SDK's interaction framework are migrated to Unity's XR Interaction Toolkit / PICO XR equivalents **[TODO: document the specific interactable/grab components replaced and any PICO-specific interaction quirks encountered]**.

5.3 Scoring and Timing System

Players earn points for correct safety decisions, quiz answers, hazard identification, and completing the game. The scoring system is implemented as a persistent Unity manager that updates a UI score display in real time **[TODO: document the scoring formula/weights and whether a timer is used]**.

5.4 Sound and Feedback

[TODO: document the audio cues and visual feedback used for correct vs. incorrect decisions, ladder/trap transitions, and Learning Card interactions]

5.5 Welcome Screen and End Screen

The application retains a welcome screen introducing the game and Junior Safety Ranger role, and a final screen summarising the player's score upon reaching the Safety Champion Zone **[TODO: insert welcome screen and end screen screenshots once the PICO 4 build is verified]**.

6 Conclusion

[TODO: reflect on the challenges faced during the Meta Quest to PICO 4 conversion (e.g. SDK incompatibilities, asset import issues, PICO 4 build/deployment setup), the solutions implemented, and suggestions for future improvement, once the conversion is complete]

7 VR Prototype

[**TODO: insert 4–6 screenshots demonstrating: the welcome screen, in-game interaction (e.g. a Good/Bad Practice decision, a Learning Card quiz), and the end screen/feedback popup, captured from the running PICO 4 build**]

8 Appendix

Table 4: Group members' contributions

Member	Detailed Contributions
Lai Yicheng (163729)	[TODO: detailed contributions]
Koay Chun Keat (164592)	[TODO: detailed contributions]
Yong Pei Yi, Angeline (163842)	[TODO: detailed contributions]
Tan Ke Jun (163204)	[TODO: detailed contributions]
Tan Yi Pin (163382)	[TODO: detailed contributions]